

January 28, 2026

Cross product (continued)

Matrices are rectangular $m \times n$ arrays of numbers

e.g. $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \in M_{2 \times 1}(\mathbb{R})$

$[3 \ 4 \ 5] \in M_{1 \times 3}(\mathbb{R})$

$\begin{bmatrix} 3 & 2 & -1 \\ 1 & 4 & 5 \end{bmatrix} \in M_{2 \times 3}(\mathbb{R})$

$\begin{bmatrix} 1 & 2 & -2 \\ 0 & \pi & \sqrt{5} \\ 3 & 0 & e^2 \end{bmatrix} \in M_{3 \times 3}(\mathbb{R})$

What do we use matrices for?

Matrices represent linear transformations:

$A \in M_{2 \times 3}(\mathbb{R})$

$\mathbb{R}^3 \longrightarrow \mathbb{R}^2$

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$

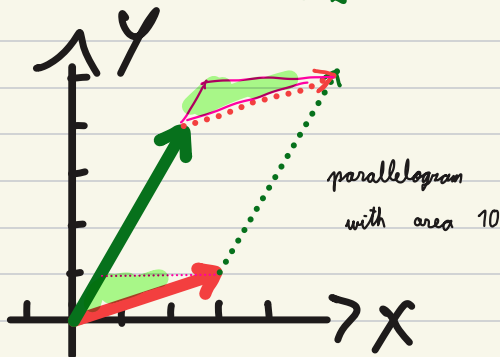
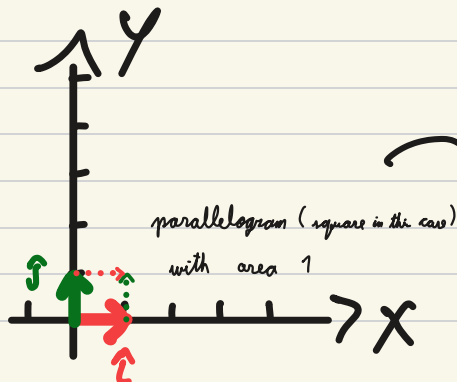
$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto A\vec{x} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$A\vec{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \end{bmatrix}$

e.g. $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$

$A\hat{i} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$A\hat{j} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$



We can measure how much a linear transformation changes the area (in 2-D), the volume (in 3-D) or the n -D volume (in n -D):

Determinant (only defined for square matrices)

in 2-D : $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

two different notations for determinant

in 3-D :

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

We will not need to go into detail and we will rarely go beyond 3-D, but... how do we get this?

Expand along a row or column, in this case along first row

$$: \begin{vmatrix} a+ & b- & c+ \\ d- & e+ & f- \\ g+ & h- & i+ \end{vmatrix} \begin{vmatrix} a+ & b- & c+ \\ d- & e+ & f- \\ g+ & h- & i+ \end{vmatrix} \begin{vmatrix} a+ & b- & c+ \\ d- & e+ & f- \\ g+ & h- & i+ \end{vmatrix}$$

we can generalize this computation to higher dimensions

$$\det(A) = +a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

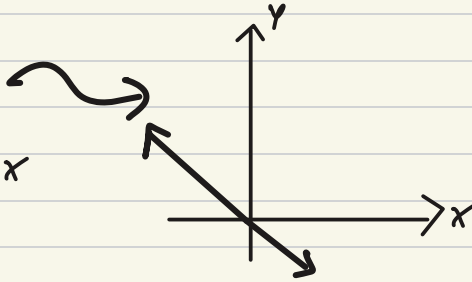
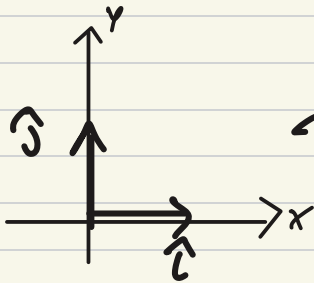
Determinants in 3-D:
Only in 3-D there is a simple mnemonic like the

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$|A| = aei + bfg + cdh - gec - hfa - idb$$

eg:

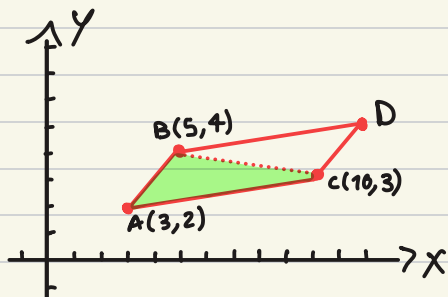
$$\begin{vmatrix} 1 & -2 \\ -1 & 2 \end{vmatrix} = (1)(2) - (-1)(-2) = 2 - 2 = 0$$



$$\det \left(\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 1 \end{bmatrix} \right) = 6$$

$$\begin{vmatrix} 1 & 2 & 0 & 1 & 2 \\ 0 & 3 & 1 & 0 & 3 \\ 2 & 1 & 1 & 2 & 1 \end{vmatrix} = 3 + 4 + 0 - 0 - 1 - 0 = 6$$

e.g. (section 1.4 example 4) Area of parallelogram and triangle



$$\vec{u} = \vec{AB} = (2, 2) \quad \vec{v} = \vec{AC} = (7, 1)$$

$$\begin{aligned} \text{Area of parallelogram } ABCD \\ &= |\det([\vec{u} \ \vec{v}])| = |\det\left[\begin{pmatrix} 2 & 7 \\ 2 & 1 \end{pmatrix}\right]| \\ &= |2 - 14| = |-12| = 12 \end{aligned}$$

(don't forget absolute value[↑] for area)

$$\text{Area of Triangle } \triangle ABC = \frac{1}{2} \|\vec{AB} \times \vec{AC}\|$$

$\vec{AB}, \vec{AC} \in \mathbb{R}^2$ - what do we mean by $\vec{AB} \times \vec{AC}$?

We see $\mathbb{R}^2$ as the x-y plane in $\mathbb{R}^3$ (i.e. $z=0$) and do cross product there

e.g. $\vec{AB} = (2, 2)$ in $\mathbb{R}^2$ see it as $(2, 2, 0)$ in $\mathbb{R}^3$

$\vec{AC} = (7, 1)$ in $\mathbb{R}^2$ see it as $(7, 1, 0)$ in $\mathbb{R}^3$

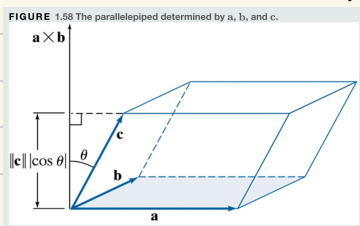
$$\vec{AB} \times \vec{AC} = (2, 2, 0) \times (7, 1, 0) = 0\hat{i} + 0\hat{j} + ((2)(1) - (2)(7))\hat{k} = -12\hat{k}$$

$$\|\vec{AB} \times \vec{AC}\| = \|-12\hat{k}\| = |-12| \|\hat{k}\| = 12$$

Another way to compute cross product:

$$\vec{a} = (a_1, a_2, a_3), \quad \vec{b} = (b_1, b_2, b_3) \quad \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

e.g. (section 1.4 example 5) Volume of a parallelepiped



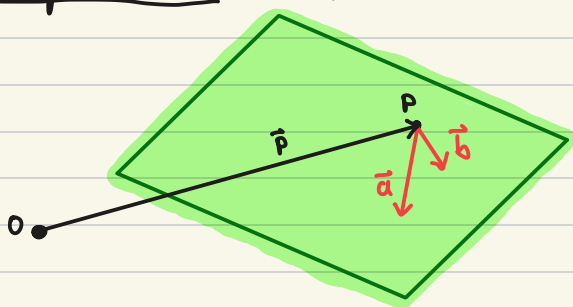
$$Vol = \left| \det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} \right| = |(\vec{a} \times \vec{b}) \cdot \vec{c}|$$

↑
triple product

1.5 Equations for planes ; Distance problems

A plane Π is defined by a point P and two non-parallel direction vectors $\vec{a}, \vec{b}$:

vector equation of a plane $\vec{r}(s,t) = \vec{p} + s\vec{a} + t\vec{b}$



We will mostly be thinking about planes inside $\mathbb{R}^3$, so

the vector equation $\vec{r}(s,t) = \vec{p} + s\vec{u} + t\vec{v}$

can be written as parametric equations of a plane inside $\mathbb{R}^3$

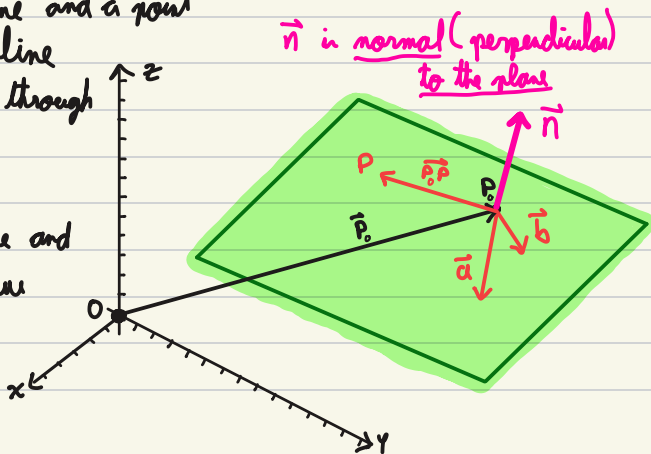
$$\begin{cases} x = c_1 + sa_1 + tb_1 \\ y = c_2 + sa_2 + tb_2 \\ z = c_3 + sa_3 + tb_3 \end{cases}$$

Important! In this section we will focus on $\mathbb{R}^3$ (1.6 will be on n -dimensional geometry)

In $\mathbb{R}^3$ if I give you a plane and a point in that plane, there is a unique line perpendicular to that plane passing through that point.

Viceversa, if in $\mathbb{R}^3$ I give you a line and a point in that line, there is a unique plane perpendicular to that line and passing through that point

plane Π : $\vec{n} \cdot (\vec{P}_0 \vec{P}) = 0$



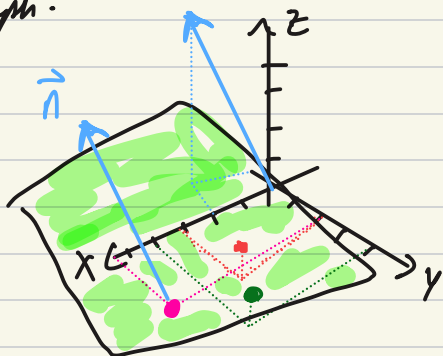
$$\vec{n} \cdot (\overrightarrow{P_0 P}) = 0 \quad P_0(x_0, y_0, z_0) \text{ some fixed point in the plane}$$

$$\left\{ \begin{array}{l} \vec{n} = (A, B, C) \\ P(x, y, z) \text{ any point in the plane} \end{array} \right.$$

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0 \quad \text{or} \quad Ax + By + Cz = D$$

e.g. (section 1.5 example 1) Plane through the point $(3, 2, 1)$
with normal vector $\vec{n} = 2\hat{i} - \hat{j} + 4\hat{k}$

standard equation of the plane: $2x - y + 4z = 8$
and graph:



$$4x - 2y + 8z = 16$$

$$P_0(3, 2, 1)$$

add 1 to x and 2 to y

$$P_1(4, 4, 1)$$

or add 2 to x and subtract 1 to z

$$P_2(5, 2, 0)$$

e.g. Find a normal vector to the plane $7x + 2y - 3z = 1$

$$\vec{n} = (7, 2, -3)$$